Constructor University Bremen

Natural Science Laboratory
Electronics

Fall Semester 2025

Lab Experiment 2 – Diode

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Place of execution	: Teaching Lab 53
Date of execution	: October 21, 2025

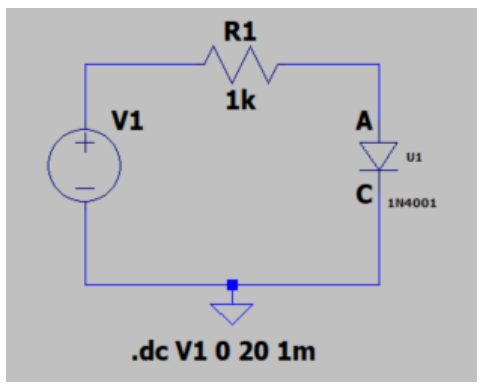
1. Introduction

The purpose of this experiment was to study the characteristics and applications of semiconductor diodes. A diode is a basic electronic component that allows current to flow in only one direction and has a nonlinear current–voltage relationship. It is made from a PN junction, formed by joining P-type and N-type semiconductor materials.

Depending on the applied voltage, the diode operates in either the forward-biased or reverse-biased region. In this experiment, the behavior of rectifier and Zener diodes was examined. Various applications such as rectifiers**,** voltage regulators**,** clampers, and clippers were studied to understand how diodes are used in shaping and controlling electrical signals.

1.1 Prelab Problem 1 – Current/Voltage characteristic of a diode

Implemented circuit using LTSpice:



Perform a DC sweep analysis:

1. Plot the diode characteristic $I_f = f(V_f)$.

2. Plot the diode characteristic $\log(I_f) = f(V_f)$.

3. Extract the values of ideality or diode factor, n, and the saturation current I_s from the graph. The voltage V_T can be assumed to be 26mV.

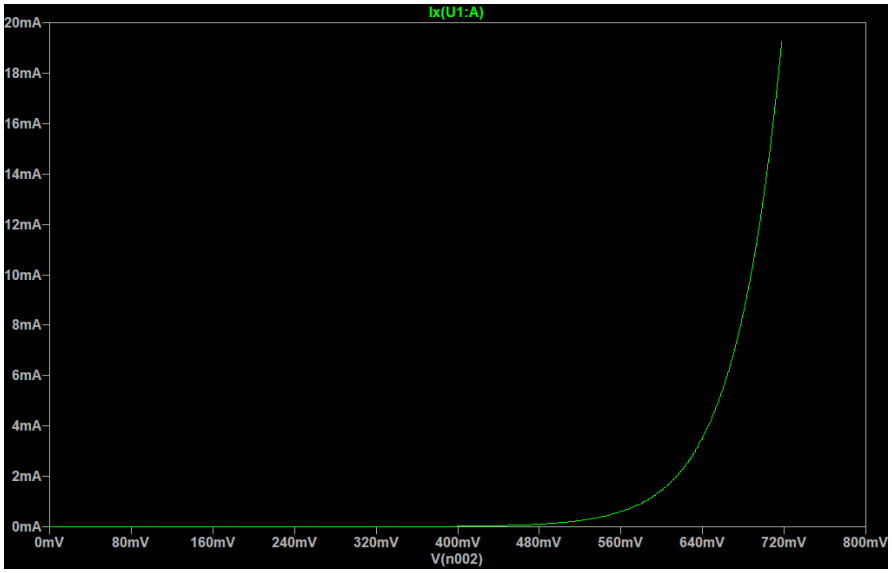


Figure 1: Diode Characteristic $I_f = f(V_f)$

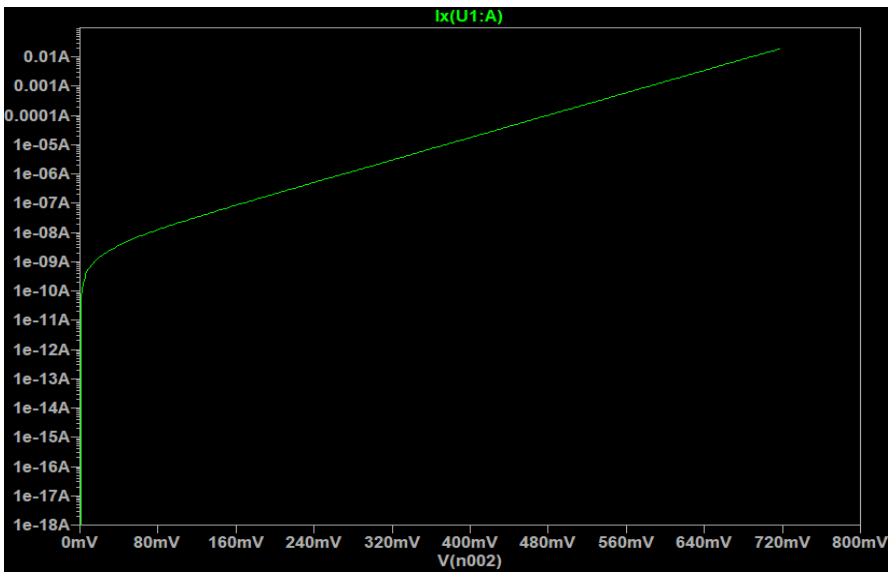


Figure 2: Diode Characteristic $\log(I_f) = f(V_f)$

1.1.1 Determination of the Ideality Factor n and I_s

The forward conduction behavior of a diode can be described by the simplified diode equation:

$$I = I_s e^{\frac{V_f}{nV_T}}$$

where I is the diode current, V_f is the forward voltage, I_s is the reverse saturation current, V_T is the thermal voltage (approximately 26 mV at room temperature), n and is the ideality factor.

Taking the natural logarithm of both sides gives:

$$\ln(I) = \ln(I_s) + \frac{V_f}{nV_T}$$

This equation represents a linear relationship of the form:

$$y = mx + b$$

Where:

- $y = ln(I)$
- $x = V_f$
- $m = \frac{1}{nV_T}$ $(the\ slope)$
- $b = ln(I_s)$ $(the\ intercept)$

By plotting $ln(I)$ versus V_f, a straight line is obtained. The slope of this line can be used to determine the ideality factor n using:

$$m = \frac{1}{nV_T} \Rightarrow n = \frac{1}{slope\ \times\ V_T} = \frac{1}{m\ \times\ V_T}$$

where m is the slope of the linear region of the $\log(I_f) = f(V_f)$ plot.

The slope m can be determined directly from the graph by selecting two points $(V_1, \ln(I_1))$ and $(V_2, \ln(I_2))$ that lie within the linear portion of the curve and applying the standard formula for the slope of a straight line:

$$m = \frac{ln(I_2) - ln(I_1)}{V_2 - V_1}$$

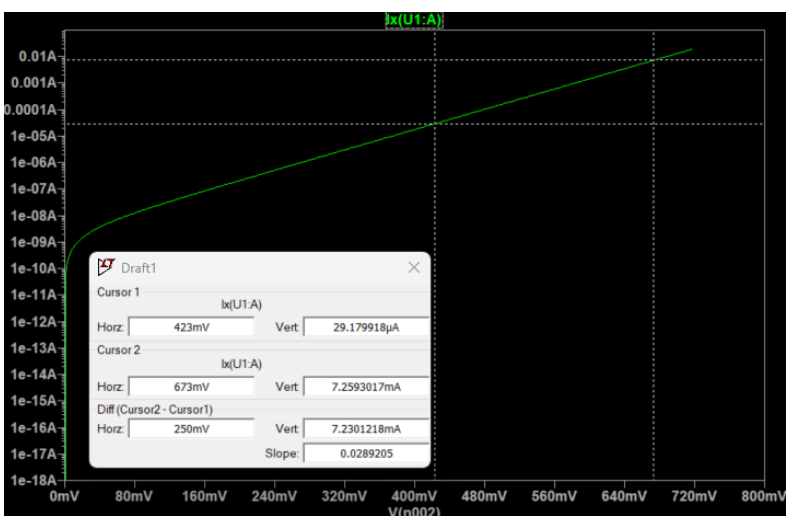


Figure 3: Taking two points from $\log(I_f) = f(V_f)$ graph using cursors

Substituting the experimental values:

$$m = \frac{ln(I_2) - \ln(I_1)}{V_2 - V_1} = \frac{ln(7.26 \times 10^{-3}) - \ln(29.18 \times 10^{-6})}{0.673 - 0.423} = \frac{5.521}{0.250} = 22.084$$

The ideality factor is then calculated using:

$$n = \frac{1}{slope \ \times \ V_T} = \frac{1}{m \ \times \ V_T}$$

Assuming a thermal voltage of $V_T = 26\ mV = 0.026\ V$:

$$n = \frac{1}{22.084 \ \times \ 0.026} = 1.75$$

$$\boxed{n = 1.75}$$

From the Diode equation I_s can be determined as:

$$I_s = I \times exp\left(-\frac{V_f}{nV_T}\right) \qquad and \qquad ln(I_s) = ln(I) - \frac{V_f}{nV_T}$$

Using the measured value for point 1 from Figure 3 and the value of calculated n:

$$\frac{V_1}{nV_T} = \frac{0.423}{1.75 \times 0.026} = 9.296$$

$$\ln(I_1) = \ln(2.918 \times 10^{-5}) = -10.442$$

$$ln(I_s) = ln(I_1) - \frac{V_1}{nV_T} = -10.442 - 9.296 = -19.738$$

$$I_s = e^{-19.738} = 2.68 \times 10^{-9}\ A$$

Using the measured value for point 2 from Figure 3 and the value of calculated n the following result was obtained:

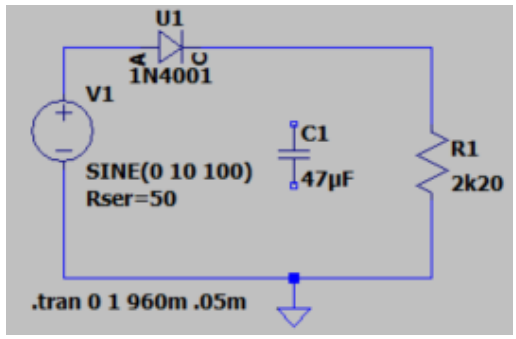
$$I_s = 2.74 \times 10^{-9}\ A$$

This value closely matches that obtained from the first data point, confirming the consistency of the measurements and the validity of the exponential diode model. Therefore, the average value of the saturation current for the 1N4001 diode can be taken as approximately:

$$\boxed{I_s = 2.72 \times 10^{-9}\ A}$$

1.2 Prelab Problem 2 – Halfwave Rectifier

Implemented circuit using LTSpice:



Perform a transient analysis (4 cycles of the sinusoidal input) for all following cases. Use the following parameters: .tran 0 1 960m .05m

1. Simulate the circuit without C1. Plot V_L, V_{in} and I_D. Measure the peak voltage at R_L and the peak current I_D. Hint: Use the cursors from the LTSpice display.

2. Simulate the circuit with C1 connected. Plot V_L, V_{in} and I_D. Measure the peak voltage at R_L, the peak current I_D, and the ripple of the voltage V_L at the load resistor. Use the formula from the handout to calculate the ripple value. Compare!

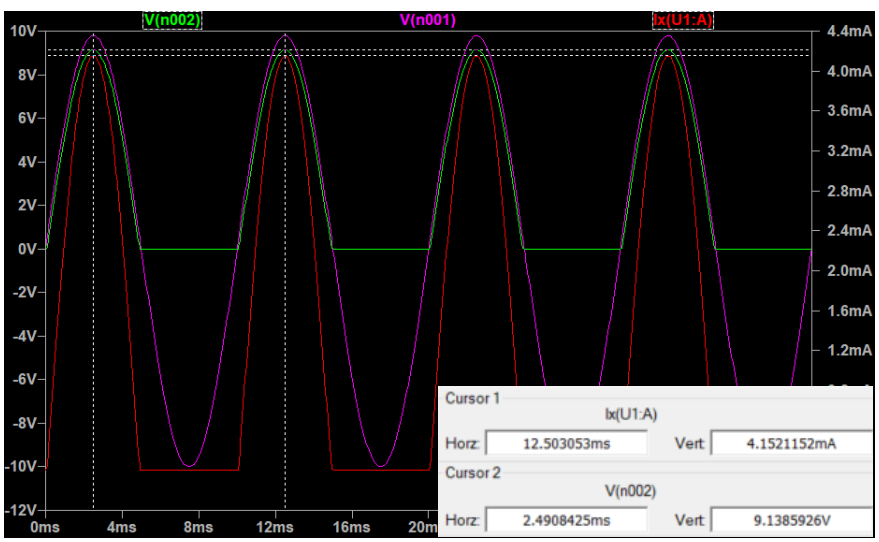
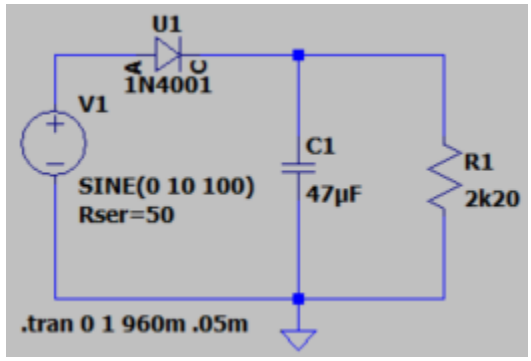


Figure 4: LTSpice Plot of I_D, V_{RL} and V_{in} without C_1 with measured the peak voltage at RL and the peak current at I_D

$$V_{RL} = 9.135\ V \quad and \quad I_D = 4.152\ mA$$

Implemented circuit using LTSpice:



The result:

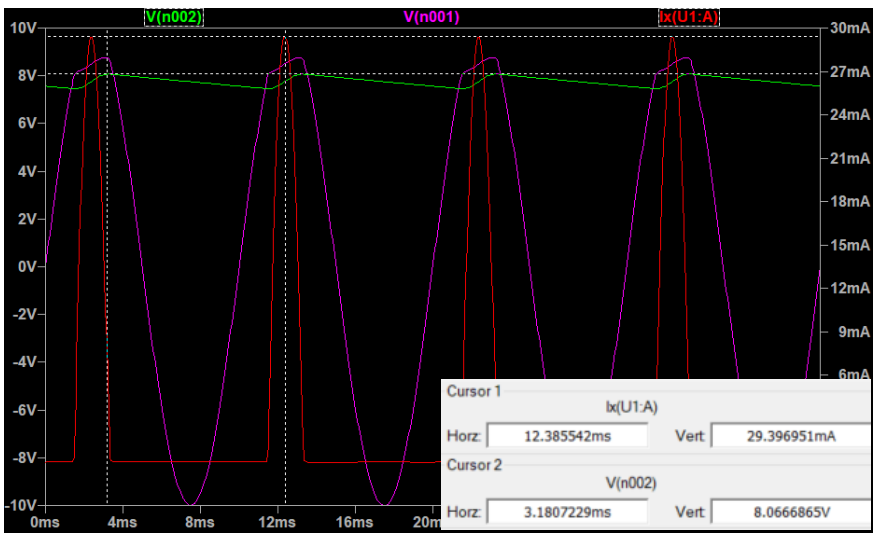


Figure 4: LTSpice Plot of I_D, V_{RL} and V_{in} with C_1 with measured the peak voltage at RL and the peak current at I_D

$$V_{RL} = 8.066\,V \quad and \quad I_D = 29.397\,mA$$

To calculate the ripple voltage at Load Resistor the plot of the ripple was zoomed in, and the maximum and minimum values of the voltage were recorded.

The ripple Voltage was calculated as:

$$V_{Ripple} = V_{max} - V_{min}$$

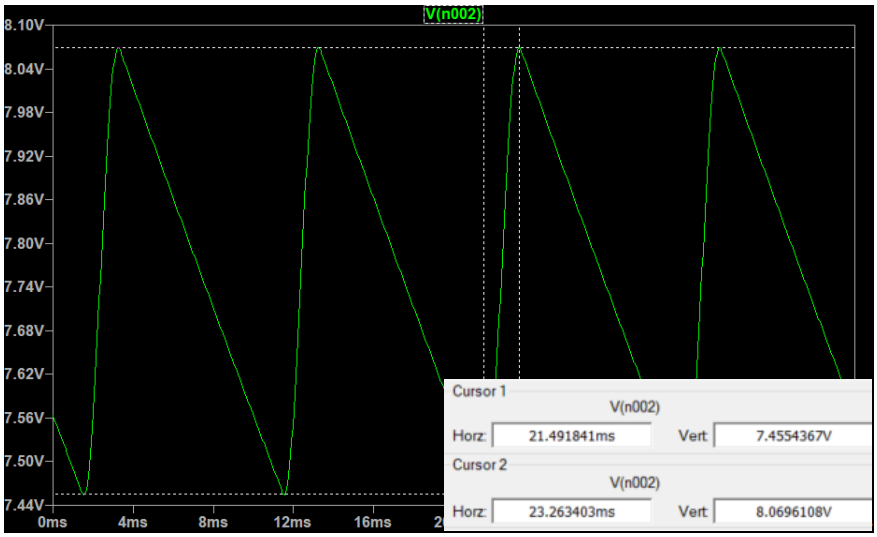


Figure 5: LTSpice Plot of V_{RL} $Ripple\ Voltage$ with measured the max and min voltage at RL

$$V_{Ripple} = V_{max} - V_{min} = 8.069 - 7.455 = 0.614\ V$$

The ripple voltage at the load resistor R_L can be calculated using the formula provided in the handout:

$$V_{Ripple} = \frac{V_P}{fCR_L}\left(1 - \sqrt[4]{\frac{R_i}{R_L}}\right)$$

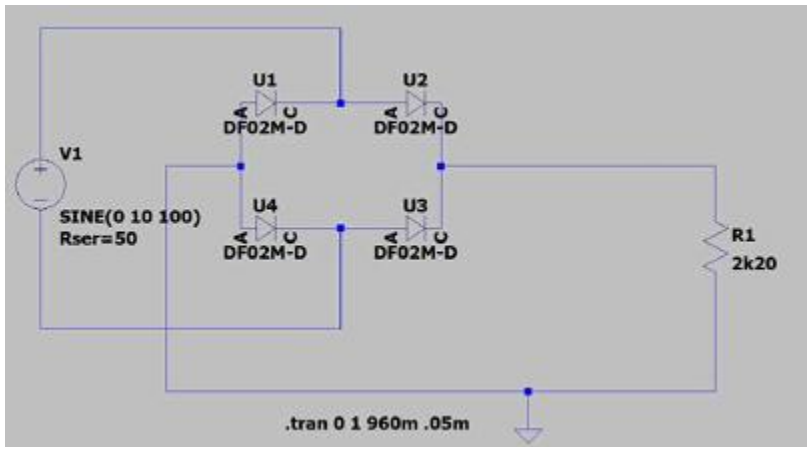
After plugging numbers:

$$V_{Ripple} = \frac{10}{100 \times 47 \times 10^{-6} \times 2200}\left(1 - \sqrt[4]{\frac{50}{2200}}\right) = 0.592\ V$$

The theoretically calculated ripple voltage at the load is $V_{Ripple} = 0.592\ V$, while the value measured from the simulation is $V_{Ripple} = 0.614\ V$ The close agreement between these values indicates that the theoretical formula provides a good approximation of the ripple, and the simulation confirms the expected behavior of the rectifier and smoothing capacitor.

1.3 Prelab Problem 3 – Fullwave rectifier

Implemented circuit using LTSpice:



Perform a transient analysis (4 cycles of the sinusoidal input) for all following cases. Use the following parameters: .tran 0 1 960m .05m

1. Simulate the circuit without C1. Plot V_L, V_{in} and I_D. Measure the peak voltage at R_L and the peak current I_D. Hint: Use the cursors from the LTSpice display.

2. Simulate the circuit with C1 connected. Plot V_L, V_{in} and I_D. Measure the peak voltage at R_L, the peak current I_D, and the ripple of the voltage V_L at the load resistor. Use the formula from the handout to calculate the ripple value. Compare!

The given circuit was simulated using LTSpice, and the waveforms of the load voltage V_L, input voltage , V_{in}, and diode current I_D were plotted to analyze the circuit's behavior and measure the ripple and peak values.

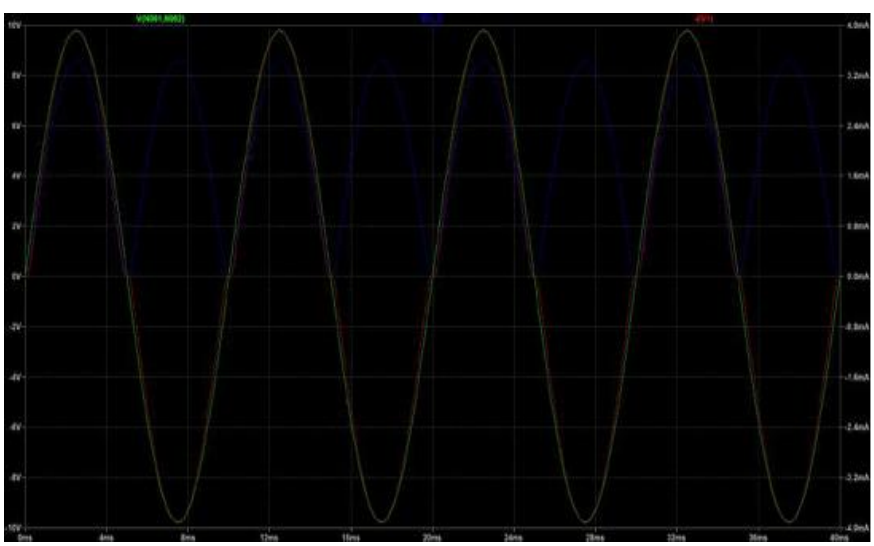
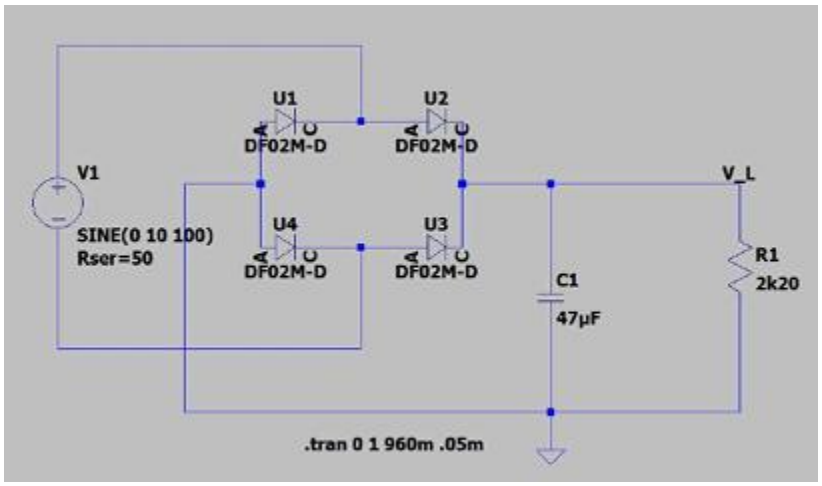


Figure 6: LTSpice Plot of I_D, V_{RL} and V_{in} without C_1 with measured the peak voltage at RL and the peak current at I_D

$$V_{RL} = 8.563\ V \quad and \quad I_D = 3.903\ mA$$

The simulation was run again with the smoothing capacitor C1 connected:



The following result was obtained:

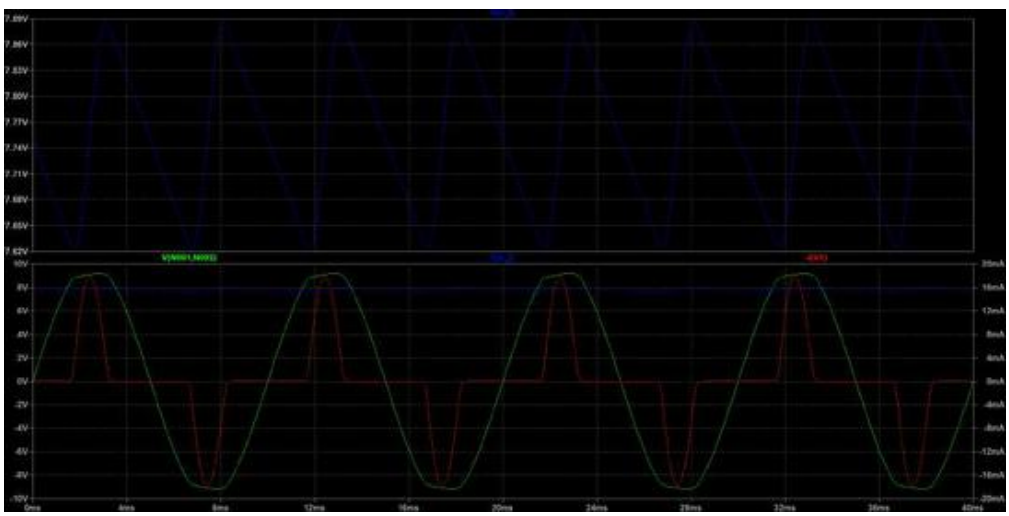


Figure 7: LTSpice Plot of $I_D, V_{RL}\ and\ V_{in}$ with C_1 with measured the peak voltage at RL, the peak current at I_D and the ripple voltage

$$V_{RL} = 7.887\ V, \quad I_D = 17.859\ mA \quad and \quad V_{Ripple} = 0.263\ V$$

The theoretical calculation for V_{Ripple}:

$$V_{Ripple} = \frac{V_P}{2fCR_L}\left(1 - \sqrt[4]{\frac{R_i}{R_L}}\right) = \frac{10}{2 \times 100 \times 47 \times 10^{-6} \times 2200}\left(1 - \sqrt[4]{\frac{50}{2200}}\right) = 0.296\ V$$

Again, the small difference between the calculated and measured ripple voltages shows that the theoretical formula accurately predicts the ripple, and the simulation validates the expected performance of the rectifier circuit with the smoothing capacitor.

1.4 Prelab Problem 4 – Fullwave rectifier

Explain the general function of the rectifier circuit with capacitor. Use simple principal circuit diagrams and the hard copies from the simulation to prove your statements!

1. What are the maximum peak voltages without a capacitor at the load for each rectifier (in our case!)? Why are these values different from the input sine amplitude? Why is there a difference between half- and full-wave rectifier?

2. Explain the differences of the current I_D for all cases. What is the consequence for the used diode in a rectifier circuit?

3. What is the influence of the ratio C ∗ RL to the quality of the output DC?

A rectifier circuit with a capacitor is used to convert an AC voltage into a DC voltage. The diodes in the circuit allow current to flow only in specific directions, while the capacitor smooths the output by charging during the voltage peaks and discharging during the valleys, reducing the voltage ripple.

From the simulation, the maximum peak voltage without a capacitor at the load is 9.132 V for the half-wave rectifier and 8.563 V for the full-wave rectifier. These values are slightly lower than the input sine amplitude because of the forward voltage drop across the diodes. In the full-wave rectifier, the current passes through two diodes per half-cycle, which reduces the peak voltage at the load compared to the half-wave rectifier. Without a capacitor, the half-wave rectifier conducts only during the positive half-cycle, producing a pulsed output voltage, whereas the full-wave rectifier conducts during both halves of the input cycle, producing pulses in each half-cycle and a higher average output voltage.

The diode current also differs depending on the rectifier type and the presence of the capacitor. Without a capacitor, the diode conducts only when forward biased, resulting in small, pulse-like

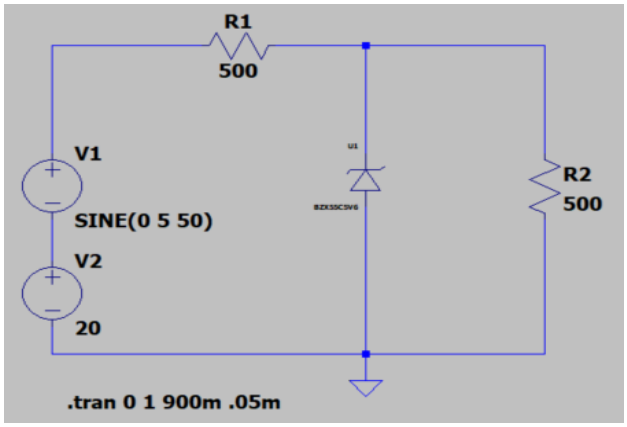
currents. When a capacitor is added, the diode conducts briefly at the voltage peaks to charge the capacitor, creating short, high-current pulses. In the simulation, the peak diode current for the half-wave rectifier increases from 4.148 mA without the capacitor to 29.397 mA with the capacitor, and for the full-wave rectifier it increases from 3.895 mA to 17.64 mA. These higher peak currents can stress the diodes, and if the current exceeds the maximum rating, it may cause overheating or damage.

The smoothness of the output voltage depends on the product of the capacitor and load resistor, C·RL. A larger C·RL value reduces the peak-to-peak ripple voltage and produces a smoother DC output, while a smaller value results in higher ripple and a more pulsating waveform. This effect is confirmed by the simulation, where adding the capacitor produces sharper voltage peaks and a more stable load voltage.

Overall, the half-wave rectifier has a slightly higher peak voltage at the load than the full-wave rectifier due to fewer diode drops. Adding a capacitor increases the peak diode current but smooths the output voltage and increasing C·RL further improves the DC quality by reducing the ripple voltage. The simulation results of the load voltage, input voltage, and diode current support these observations.

1.5 Prelab Problem 5 – Zener Diode

Implemented circuit using LTSpice:



1. Perform a transient analysis (5 cycles of the sinusoidal input). Plot the input voltage (DC + AC voltage) and the output voltage across the load resistor RL.

2. Explain the operation of the circuit.

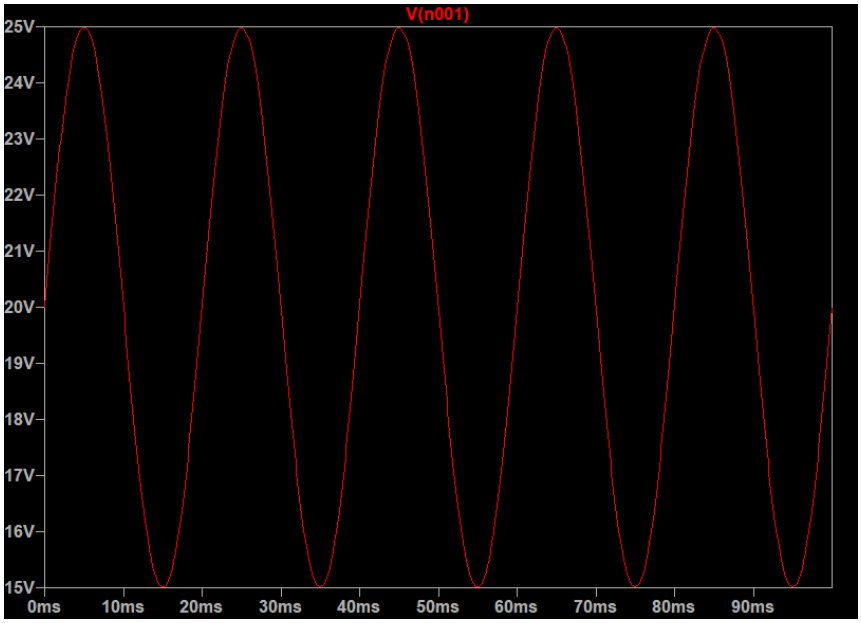


Figure 8: LTSpice Plot of V_{in}

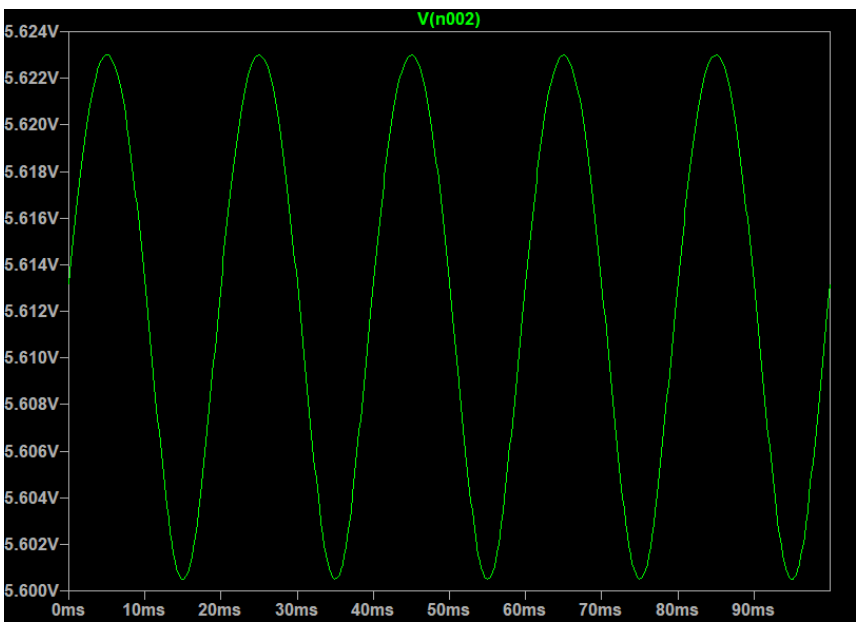


Figure 9: LTSpice Plot of V_{RL}

The circuit functions as a voltage regulator, designed to maintain a stable output voltage regardless of variations in the input voltage or changes in the load current. When the voltage at the load rises above the desired level, the Zener diode conducts more current. This increased current flows through the series resistor, causing a larger voltage drop across it and reducing the voltage applied to the load, thereby keeping the load voltage constant. Conversely, if the load voltage drops, the Zener diode conducts less current, decreasing the voltage drop across the resistor and allowing the

load voltage to rise back to its regulated value. In this way, the Zener diode and series resistor work together to stabilize the voltage across the load under varying input or load conditions.

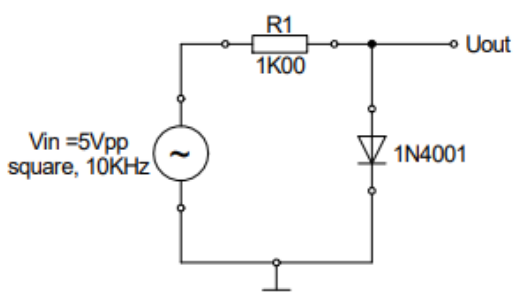
2. Execution – Set-up and Results

Tools and Equipment

- 47µF ,100µF and 5 x 1µF Capacitors
- 1N4001, DF02M-E rectifier, BZX85C5V6 Zener and 4 x 1N4148 Diodes
- 1kΩ, 100kΩ and 2K20Ω resistors
- Breadboard
- Oscilloscope for measurement
- Function Signal Generator

2.1 Execution Problem 1 – Diode Switching Characteristic

The objective of this part of the experiment was to investigate the reverse-to-forward and forward-to-reverse transition behavior of a rectifier diode and a signal diode**.** The required circuit was assembled on a breadboard according to the given circuit diagram. The diodes were connected as shown, and appropriate input voltages were applied to observe the diode's transition between forward and reverse bias conditions. The corresponding waveforms and voltage changes were then observed and recorded.



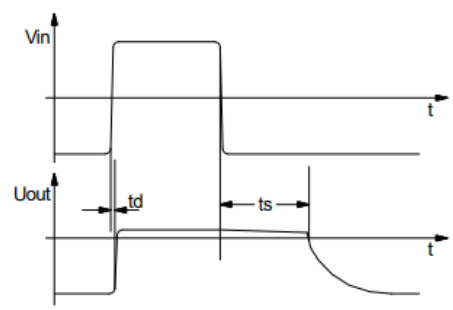


Figure 10: Determination of ts and td

Using the following figure as a reference, the storage time (ts) and transition or delay time (td) of the 1N4001 rectifier and the 1N4148 signal diode were measured:

Figure 11: td for 1N4001 rectifier diode

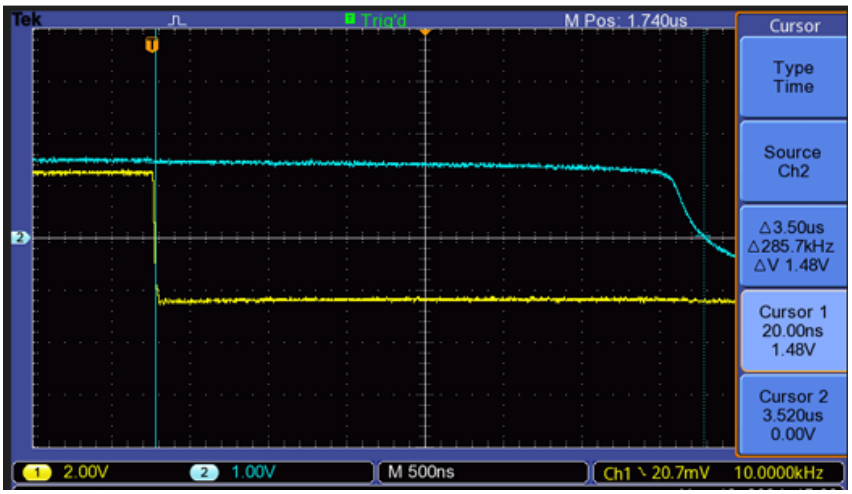


Figure 12: ts for 1N4001 rectifier diode

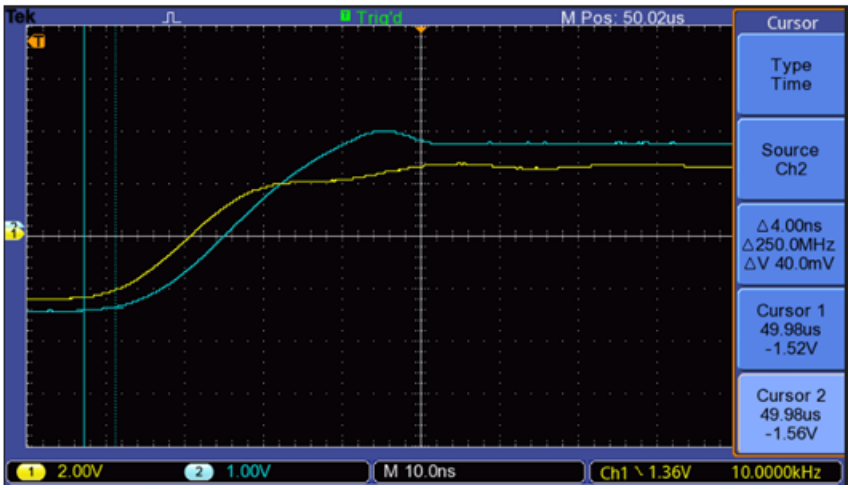


Figure 13: td for 1N4148 signal diode

Figure 14: ts for 1N4148 signal diode

According to the obtained results, the following table was prepared:

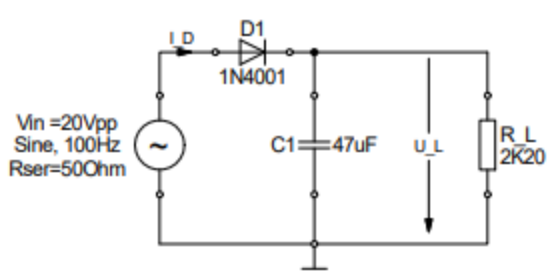
Diode	t_d	t_s
1N4001	11.6 ns	3.5 us
1N4148	4 ns	8.8 ns

Table 1: A storage and delay times for
1N4001 and 1N4148 diodes

2.2 Execution Problem 2 – Rectifier

2.2.1 Half-Wave rectifier

For the first part of the second experiment, the following circuit was constructed on the breadboard:



Capacitor C1 was removed, and the peak voltages of Vin and VL were measured using the measure function of the oscilloscope. The measured values can be seeing on the figures.

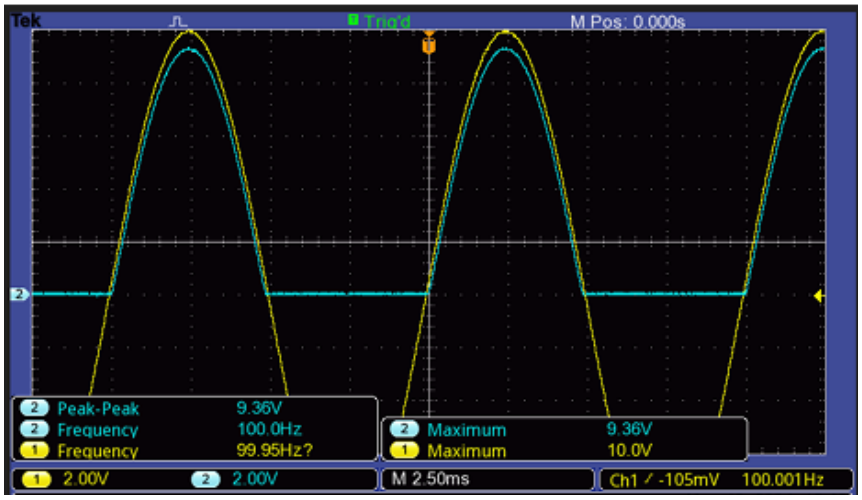


Figure 15: Half-Wave rectifier – without C1 – measured values

Capacitor C1 was then inserted into the circuit, and the measurements were taken again.

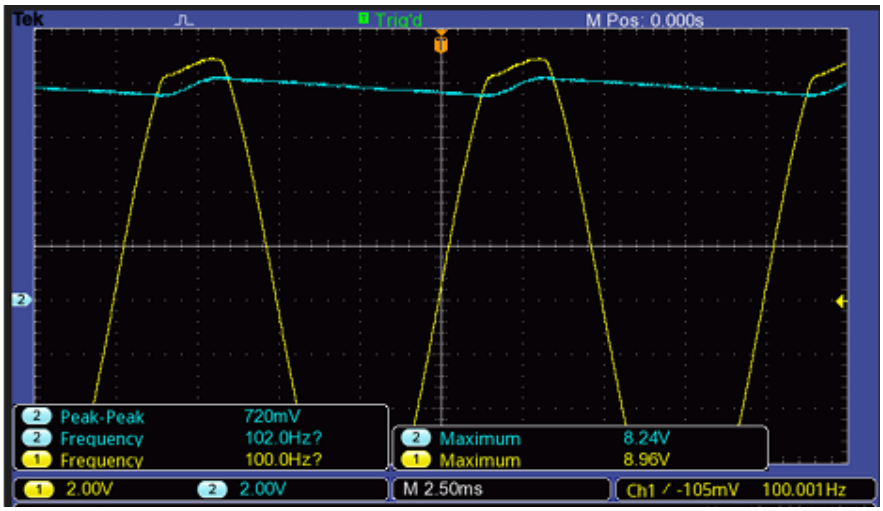


Figure 16: Half-Wave rectifier – with C1 – measured values

The ripple voltage signal (VL) was zoomed in, and its peak-to-peak value was measured, as shown in Figure below AC coupling was selected in the channel menu of the oscilloscope.

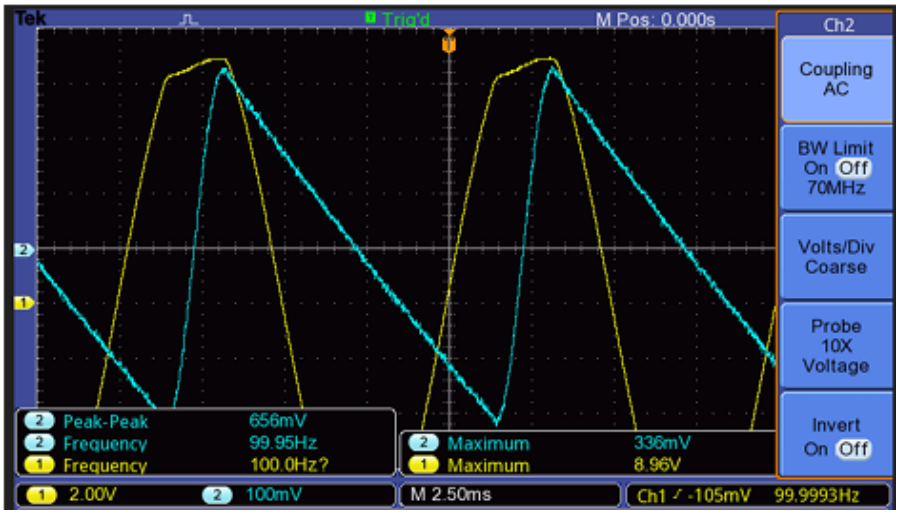
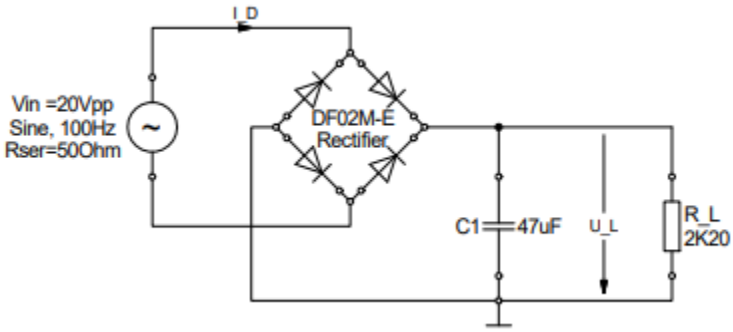


Figure 17: Zoomed in signal of the Ripple Voltage with measured value

2.2.2 Full-Wave rectifier

For the second part of the second experiment, the following circuit was constructed on the breadboard:



Capacitor C1 was removed, and the peak voltages of Vin and VL were measured using the measure function of the oscilloscope. The measured values can be seeing on the figures.

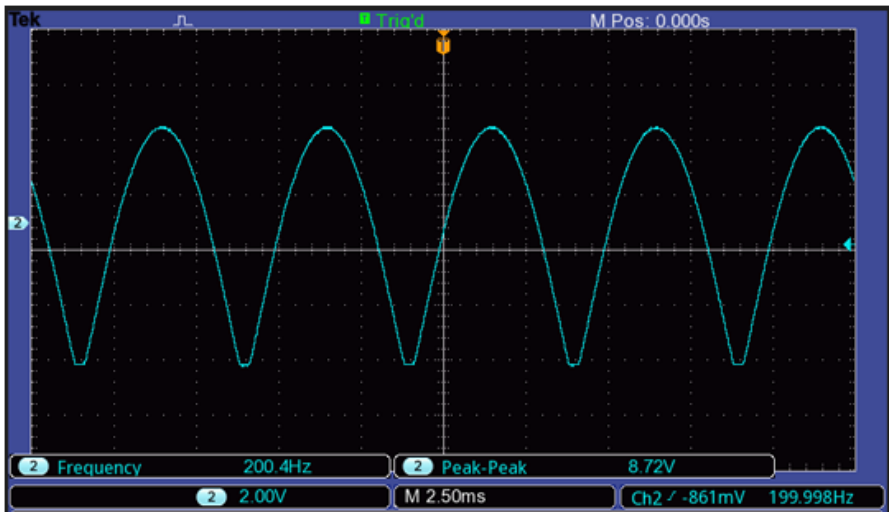


Figure 18: Full-Waved rectifier – Without C1 – measured values

Capacitor C1 was then inserted into the circuit, and the measurements were taken again.

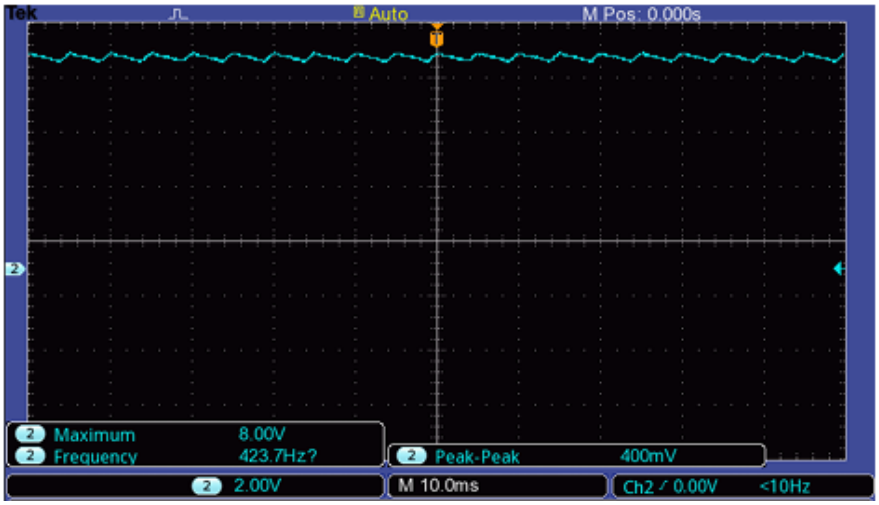


Figure 19: Full-Waved rectifier – With C1 – measured values

The ripple voltage signal (VL) was zoomed in, and its peak-to-peak value was measured, as shown in Figure below AC coupling was selected in the channel menu of the oscilloscope.

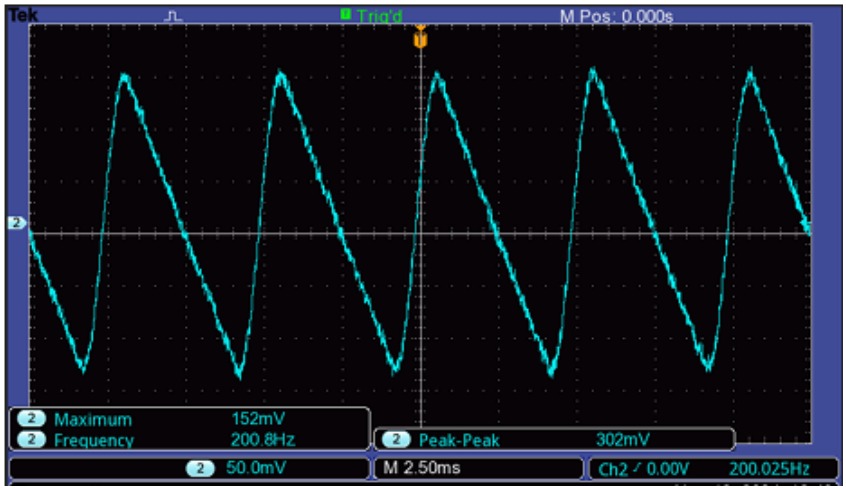
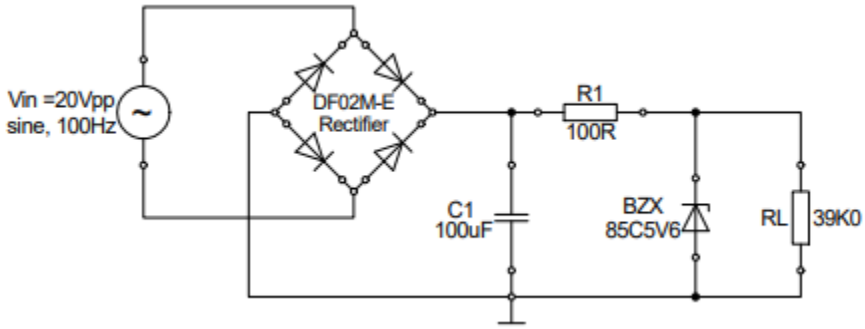


Figure 20: Zoomed in signal of the Ripple voltage with measured values

2.3 Execution Problem 3 – Zener Diode

For the third part of the experiment, the following circuit was implemented.



The oscilloscope was then used to measure the DC and ripple voltages at C1, as well as the output DC voltage and ripple voltage across RL. Hard copies of the waveforms were taken along with the recorded measurements.

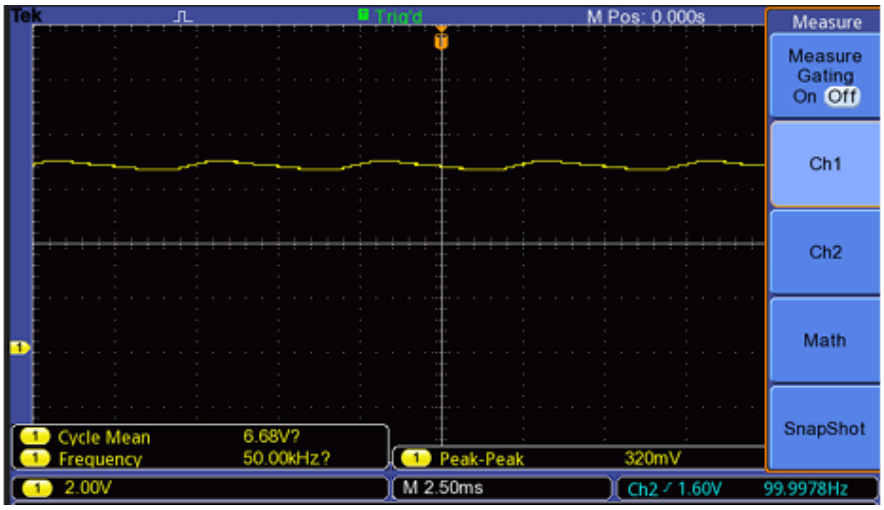


Figure 21: DC Voltage at C1

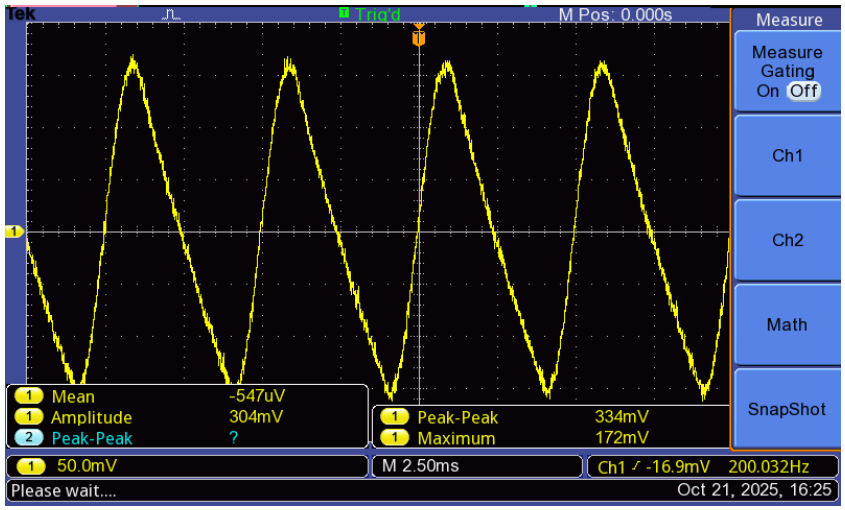


Figure 22: Ripple Voltage at C1

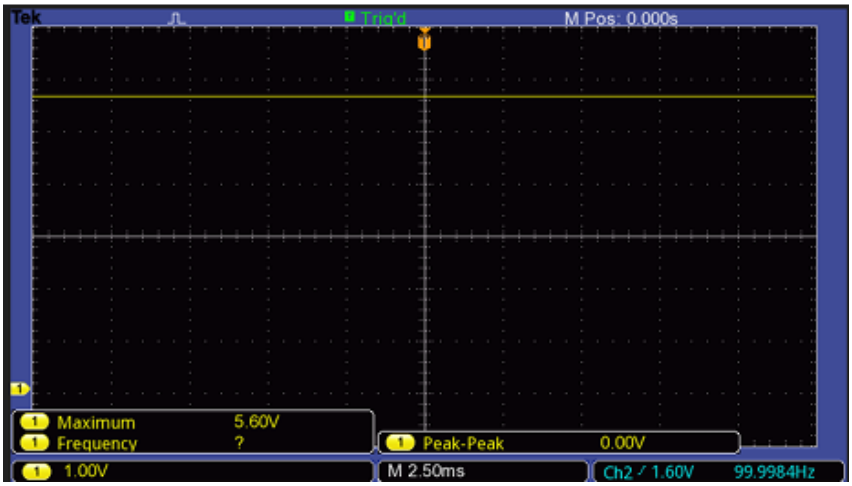


Figure 23: DC Voltage at RL

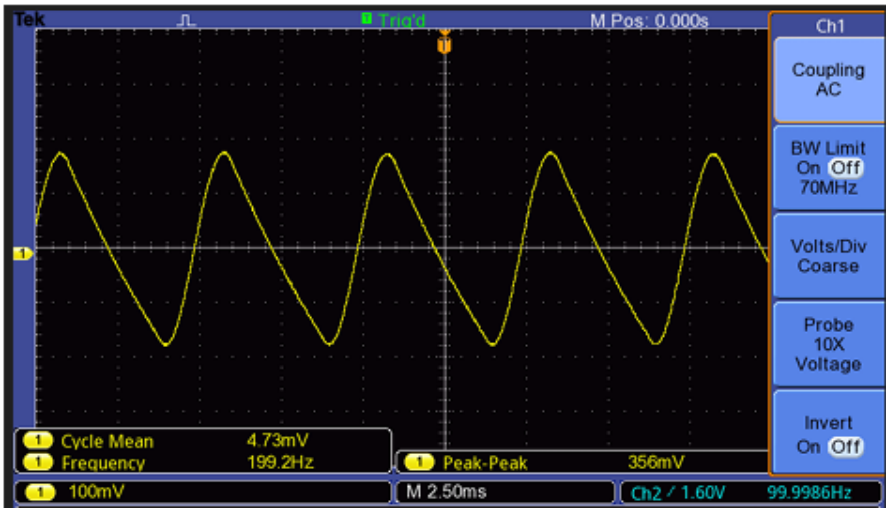
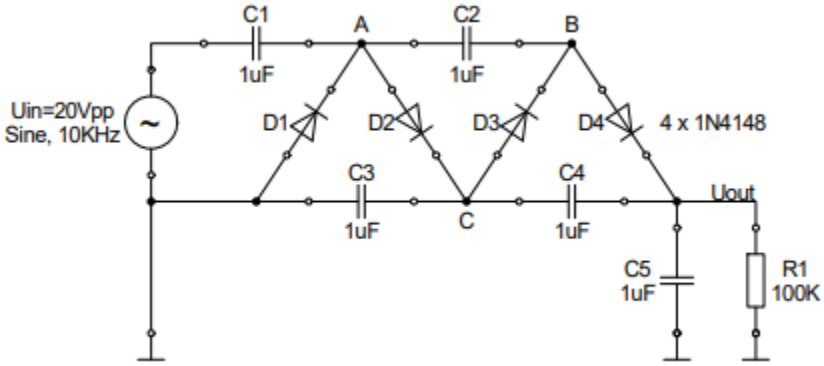


Figure 24: Ripple Voltage at RL

2.4 Execution Problem 4 – Voltage Multiplier

In the last part of the experiment, the circuit was set up on the breadboard for testing.



The oscilloscope was then used to record the voltages at points A and C, as well as at points B and Uout. The average values were obtained using the measure function, and hard copies of the waveforms were collected.

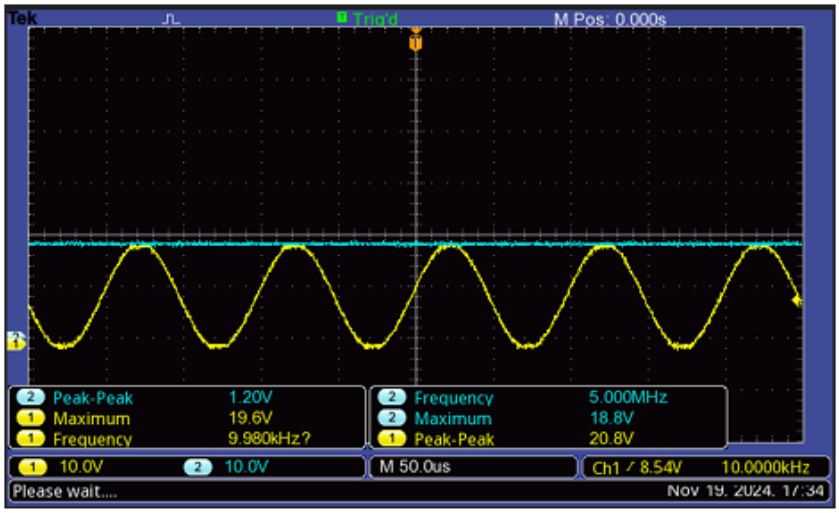


Figure 25: Voltage at ‘A’ and ‘C’

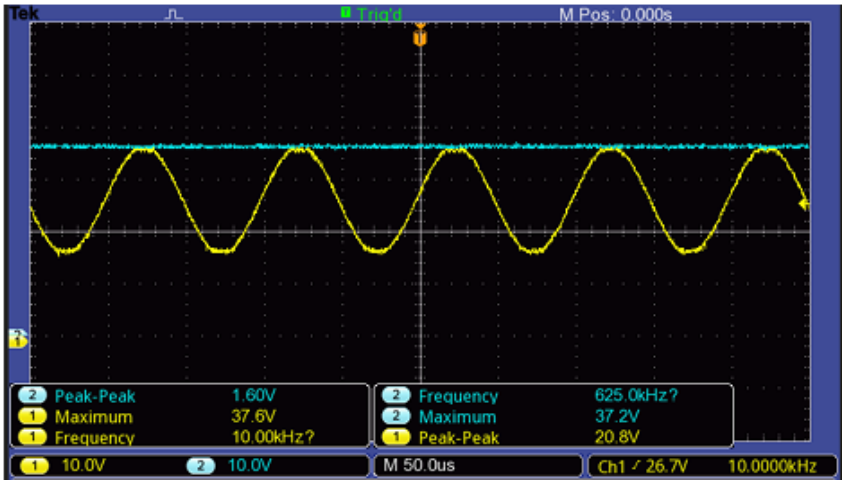


Figure 26: Voltage at ‘B’ and ‘Uout’

Similarly, the ripple voltage at Uout was measured. The average function in the acquire mode of the oscilloscope was used to smooth the waveform and obtain a more accurate value.

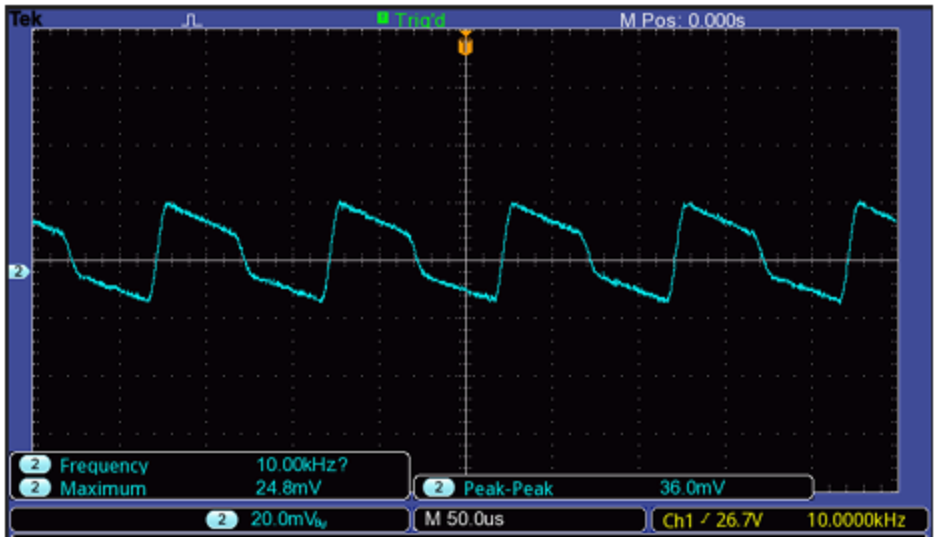


Figure 27: Ripple Voltage at 'Uout'

Finally, a multimeter was used to measure the voltages across 'C' and 'Uout' and Based on these results, the following table was prepared.

Voltage 'A'	19.6 V
Voltage 'C'	18.8 V
Voltage 'B'	37.6 V
Voltage 'Uout'	37.2 V
Ripple 'Uout'	36 mV
Multimeter 'C'	17.712 V
Multimeter 'Uout'	35.473 V

Table 2: Measured Values for Problem 4

3. Evaluation – Diode

3.1 Evaluation Problem 1 – Diode switching characteristics

The storage times measured were 4.16 µs for the 1N4001 and 12.6 ns for the 1N4148. The storage time of the 1N4001 was therefore much longer than that of the 1N4148. This behavior was caused by the larger amount of charge that was stored in the junction of the 1N4001 while it was forward-biased. When the polarity was reversed, time was required for the stored minority carriers to recombine or to be removed from the depletion region; this delay defines the diode's storage (reverse recovery) time. The larger physical junction area and the device design of the 1N4001

(lower doping and larger junction volume) led to greater charge storage and a correspondingly longer switching time, whereas the 1N4148 was designed with a small junction and higher switching speed, so only a very small charge had to be removed and a much shorter storage time resulted.

The long storage time of the 1N4001 imposes clear limits in high-frequency applications. For example, at 100 kHz the signal period is 10 µs: the 1N4001 storage time of 4.16 µs corresponds to about 41.6% of one period, and at 200 kHz (period 5 µs) it corresponds to about 83.2% of one period. Because of this, the 1N4001 would not reliably follow rapid polarity changes and would introduce pulse distortion, delayed switching, and attenuation of high-frequency components. In AM demodulation and other signal processing tasks operating at several 100 kHz, such distortions would degrade the recovered signal and increase error.

By contrast, the 1N4148 storage time of 12.6 ns is negligible compared with periods at 100–200 kHz (it represents ~0.13% and ~0.25% of a 10 µs and 5 µs period respectively). Therefore, the 1N4148 can be expected to switch cleanly at those frequencies and to introduce minimal distortion. However, the 1N4148 cannot safely replace the 1N4001 in power applications because it is not rated for the same high currents and repetitive power dissipation.

In summary, the 1N4001 was found suitable for low-frequency or power-rectification tasks where switching speed is not critical, whereas the 1N4148 was found suitable for high-frequency signal applications (such as AM demodulation), where fast recovery is required but high current capability is not.

3.2 Evaluation Problem 2 – Rectifier

1. Input Signal → 2. Transformer → 3. Rectifier Circuit → 4. Filter Circuit → 5. Voltage Regulator → 6. Regulated DC output

- **Input Signal:**
The input signal is the AC voltage supplied from the mains, which serves as the source to be converted into DC.

- **Transformer:**
A transformer is used to step up or step down the AC input voltage to the desired level. It also provides electrical isolation between the mains supply and the circuit for safety.

- **Rectifier Circuit:**
The rectifier converts the AC voltage from the transformer into a pulsating DC voltage. In a full-wave rectifier, both halves of the AC waveform are used, which reduces the ripple content compared to a half-wave rectifier.

- **Filter Circuit:**

The filter smooths the pulsating DC signal by reducing the ripple voltage. This is achieved by using capacitors that store and release charge, making the output voltage more constant.

- **Voltage Regulator:**

The voltage regulator maintains a steady DC output voltage even when the input voltage or load current changes. A Zener diode or a regulator IC is often used for this purpose.

Vripple, V	Half-wave rectifier	Full-wave rectifier
Simulated	0.614 V	0.263 V
Calculated	0.5916 V	0.296 V
Measured	0.656 V	0.320 V

Table 3: Comparison of ripple voltage values

From the obtained results, it was observed that the calculated, simulated, and measured values were all within a similar range, with only small deviations present. These differences can be attributed to several sources of error. Instrumental errors from the oscilloscope and function generator may have affected the accuracy of the measurements. Additionally, variations in component tolerances, such as the resistance and capacitance values, and the non-ideal behavior of the diodes, could have introduced small voltage drops and discrepancies.

The diodes used in the practical setup might also have shown slight performance differences compared to the ideal models used in the simulation, due to factors such as junction degradation, material imperfections, or temperature variations. Furthermore, minor inaccuracies may have been introduced while using the cursor or measurement functions in the LTspice simulation software. Altogether, these factors contributed to the small but noticeable variations between the theoretical, simulated, and experimental results.

3.3 Evaluation Problem 3 – Zener Diode

To determine the current flowing through the Zener diode, Kirchhoff's Current Law (KCL) was applied. First, the current through the load resistor was calculated using the measured values from Section 2.3:

$$I_{RL} = \frac{5.6}{39000} = 1.436 * 10^{-4}\ A$$

Similarly, the current through resistor R_1 was found as:

$$I_{R1} = \frac{6.88 - 5.6}{100} = 0.0128\ A$$

By applying KCL at the node connecting the resistors and the Zener diode:

$$I_Z = I_{R1} - I_{RL} = 0.0128 - 1.436 * 10^{-4} = 0.01266\ A$$

Thus, the current through the Zener diode was calculated to be 12.66 mA**.**

3.4 Evaluation Problem 4 – Voltage Multiplier

When comparing the circuit used in part four of the experiment, it was observed that the configuration resembled the clamper circuit described in the diode applications section of the handout. The circuit operated as a voltage multiplier, producing a DC output voltage that was significantly higher than the peak value of the input AC signal.

During the positive half-cycle of the input waveform, the capacitor C1 was charged up to the peak value of the input voltage and retained this charge. In the following negative half-cycle, the voltage across C1 added to the input AC voltage, effectively doubling the potential at point A. This allowed diode D2 to conduct, charging capacitor C3 with the doubled voltage.

In the next stage, during the subsequent positive half-cycle, the voltage stored in C3 was used to charge C2 through diode D3. Then, during the negative half-cycle, the voltage across C2 transferred charge to C4 through diode D4. As a result, the voltage at point B reached approximately four times the input peak voltage, with C4 maintaining the stored charge. Finally, capacitor C5 acted as a filter to minimize any remaining ripple, producing a smoother DC output voltage across the load.

The Circuit was constructed in LT:

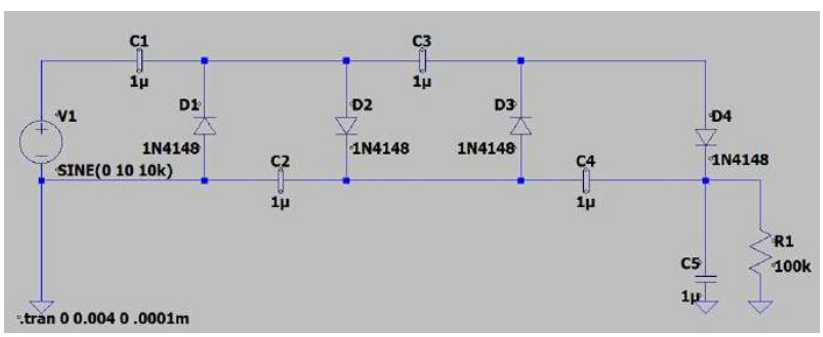


Figure 28: LTspice Circuit for Problem 4

By plotting V_{out} and V_{in} and measuring their peak magnitudes using the cursor function, the values were obtained. The time base was adjusted to start from 3 ms to ensure that the output waveform had reached a stable peak before measurement.

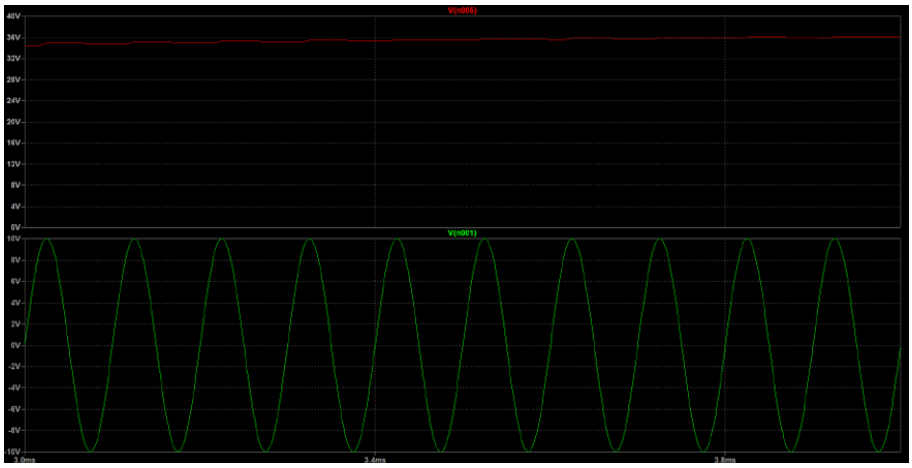


Figure 29: LTspice Plot of V_{out} and V_{in}

The output voltage (V_{out}) was measured to be 36.0129 V, while the input voltage (Vin) was measured as 9.998 V, which is approximately 10 V, as expected. Therefore, for the ideal case, the multiplication factor (m) between the input amplitude and the output voltage was calculated as:

$$m = \frac{Output}{Input} = \frac{36.0129}{10} = 3.601$$

Similarly, using the measured values obtained in Section 2.4, the multiplication factor was calculated as:

$$m = \frac{Output}{Input} = \frac{36.4}{10} = 3.64$$

The difference between the two results occurred because the diodes used in the LTspice simulation were modeled under ideal conditions, while the diodes used in the physical experiment were affected by real-world factors such as material degradation, temperature variations, and manufacturing tolerances. In addition, practical effects like capacitor leakage and the presence of load resistance caused a slight reduction in the overall DC output voltage.

The circuit can be divided into two main sections. In the first section, from node A to C, capacitor C3 stored a voltage approximately twice the input amplitude. In the second section, from node C to B, capacitor C4 stored a voltage about four times the input amplitude. To achieve a higher output voltage, the capacitance values can be increased, as larger capacitors have greater charge storage capacity and produce higher and more stable DC output levels.

The circuit was simulated again in LTspice using an input frequency of 100 Hz:

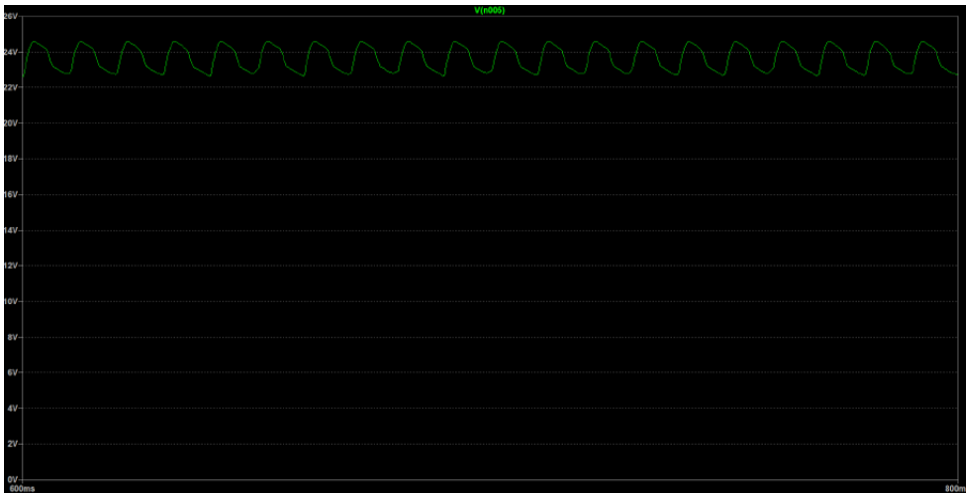


Figure 30: LTspice Plot of V_{out} and V_{in} for 100 Hz

It was observed that when the input frequency was reduced to 100 Hz, the output voltage decreased to 24.58 V. The ripple voltage increased noticeably, and the output waveform became a pulsating DC signal rather than a smooth DC output.

4. Evaluation – Diode

Throughout the experiment, the characteristics, and applications of different types of diodes were investigated. In the first part, the switching behavior of two diodes, 1N4001 and 1N4148, was examined. It was found that the 1N4001 diode had a longer storage time, making it suitable for low-frequency and power rectification applications, while the 1N4148 diode had a much shorter switching time, making it more appropriate for high-frequency signal applications such as AM demodulation.

In the following parts, half-wave and full-wave rectifier circuits were constructed and tested with and without filter capacitors. It was observed that the addition of a capacitor significantly reduced the ripple, producing a smoother DC output. The measured, simulated, and calculated values were compared and found to be within a close range, with small deviations caused by non-ideal component behavior and measurement inaccuracies.

The Zener diode circuit was then analyzed, and it was confirmed that the diode functioned as a voltage regulator by maintaining a nearly constant output voltage despite variations in input or load conditions. In the final part, a voltage multiplier circuit was implemented using diode–capacitor stages. The circuit successfully produced a higher DC output voltage, demonstrating the principle of voltage multiplication. It was also observed that a reduction in input frequency increased the ripple and reduced the output voltage.

Overall, the experiment demonstrated that diodes are versatile components with a wide range of applications in power conversion and signal processing. Minor differences between theoretical,

simulated, and measured results were mainly caused by instrumental inaccuracies, component tolerances, temperature effects, and non-ideal diode behavior. Despite these factors, the experimental outcomes closely matched the expected theoretical performance.

5. References

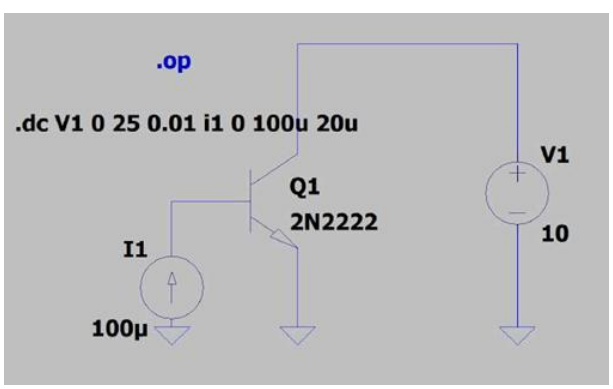
- Behzad, Razavi, 2nd edition, Fundamentals of Microelectronics
- 20240829-co-526-b manual- U.Pagel, Page(11-39)

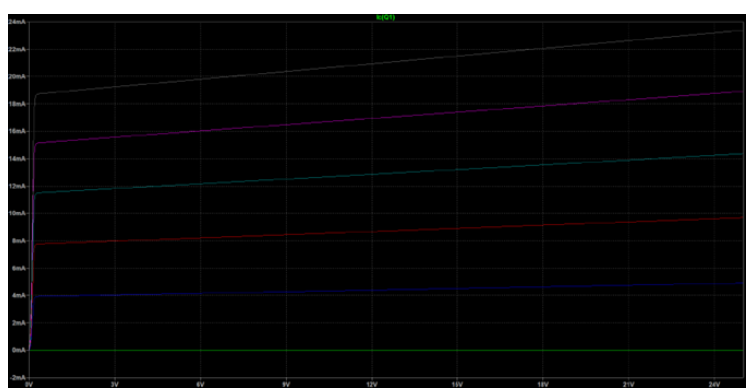
6. Appendix

6.1 Problem 1 – Simulate a circuit using a Natelist

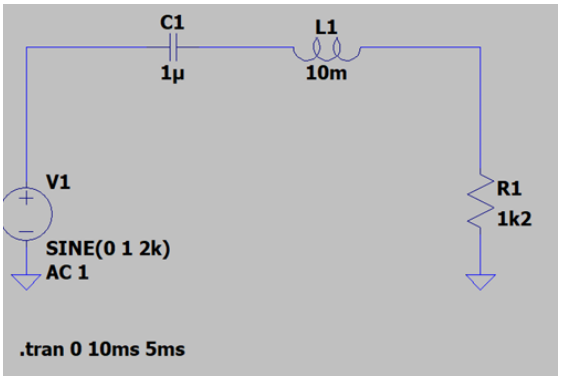
```
V1 1 0 5

Rr1 1 2  11k

Rr2 2 0 1k

Rr3  1 3 1e3

Rr4 3 0 1000

Rr5 2 3 1k

.op

.en
```

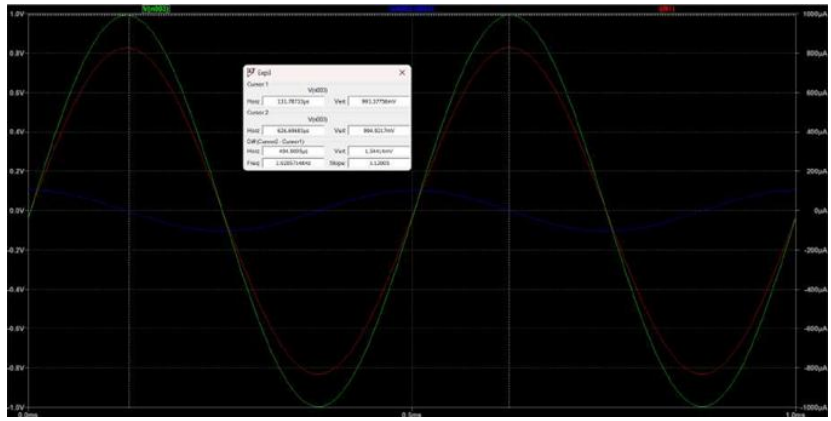
6.2 Problem 2 – Perform a DC Sweep Simulation





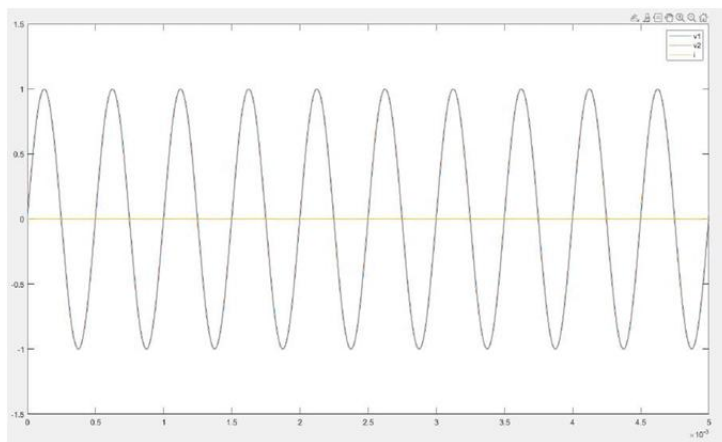
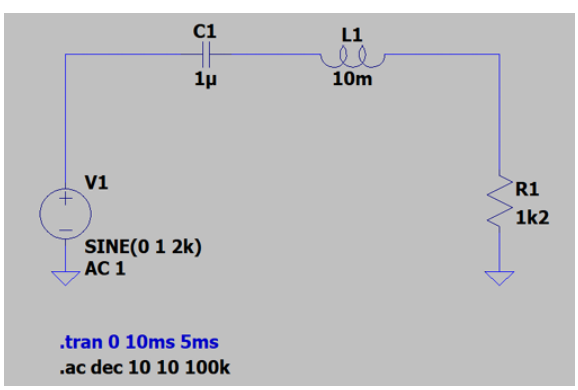
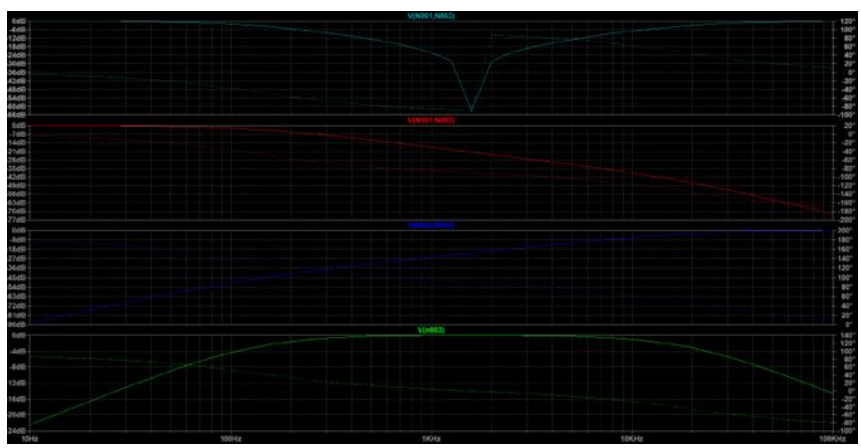
6.3 Problem 3 – Perform a Transient Analysis





6.4 Problem 4 – Export Data to Matlab

```
% 1. Clear the MatLab environment
clear; clc; close all
% 2. Read the data from the LTSpice file.

y_arr = readmatrix('Exp3.txt');
% 3. Read every column of the matrix into a separate array
    for time,
% voltage 1, and voltage 2 and the current
t = y_arr(1:end, 1); % start reading column 2 (first is
   empty!)
v1 = y_arr(1:end, 2);
v2 = y_arr(1:end, 3);
i = y_arr(1:end, 4);
% 4. Plot the different properties
plot (t, v1);
hold on
plot (t, v2);
hold on
plot (t, i);
legend('v1','v2','i');
```



6.5 Problem 5 – Perform a AC Analysis